Chapter 1: Overview and Descriptive Statistics

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Outline

1. Some basic concepts
2. Describing Data with Graphs
   - Histogram
   - Stem-and-leaf plots and scatter plots
3. Describing Quantitative Data
   - Measures of Central Tendency
   - Measures of variation

Population and sample

- **Population:** the set of all elements bearing the characteristic of interest or under study.
- **Sample:** a subset selected from the population.
- **Parameter:** a specific characteristic of a population.
- **Statistic:** a specific characteristic of a sample.

Examples of a population

- The number of registered voters
- The income of households in a city
- The grade point average of all students in a university
- The weight of products in a factory

Typically, we cannot select all elements of the population to study because:

- The number of elements in the population is very large
- Time and cost constraints do not permit it
- It may damage the elements of the population

Therefore, we only conduct research on samples selected from the population.

Random Sampling

A random sample consisting of n elements selected from a population must satisfy the following conditions:

- Each element in the population must be selected randomly and independently
- Each element in the population has the same likelihood of being selected (equal probability of selection)
- Every sample of size n also has the same likelihood of being selected from the population

The method of simple random sampling:

+ Number the elements of the population from 1 to N. Prepare slips also numbered in this way.
+ Mix the slips well, then select n slips with replacement. The elements of the population corresponding to the numbers on the drawn slips will be selected as the sample.

Descriptive Statistics

- **Descriptive statistics:** is the process of collecting, summarizing, and processing data to transform data into information.
  - Data collection: surveys, measurements, ...
  - Data presentation: using tables and graphs
  - Data summarization: calculating sample statistics such as sample mean, sample variance, median, ...

Inferential Statistics

- **Inference** is a process of drawing conclusions or making decisions about a population based on research results from a sample.
- **Inferential statistics:** processes information obtained from descriptive statistics, thereby providing the basis for predictions, forecasts, and estimations.
  - Estimation: for example, estimating the proportion of defective products in a factory, or estimating the proportion of passengers who purchased tickets but were absent on a flight.
  - Hypothesis testing: for example, testing the claim that the average profit of a store in a month is 300 million.

Introduction

- Describing data using graphs provides an overview of the data before entering into detailed analysis.
- The types of graphs used will depend on the type of variable to be analyzed.
- In this section, we will primarily examine histograms and stem-and-leaf plots.

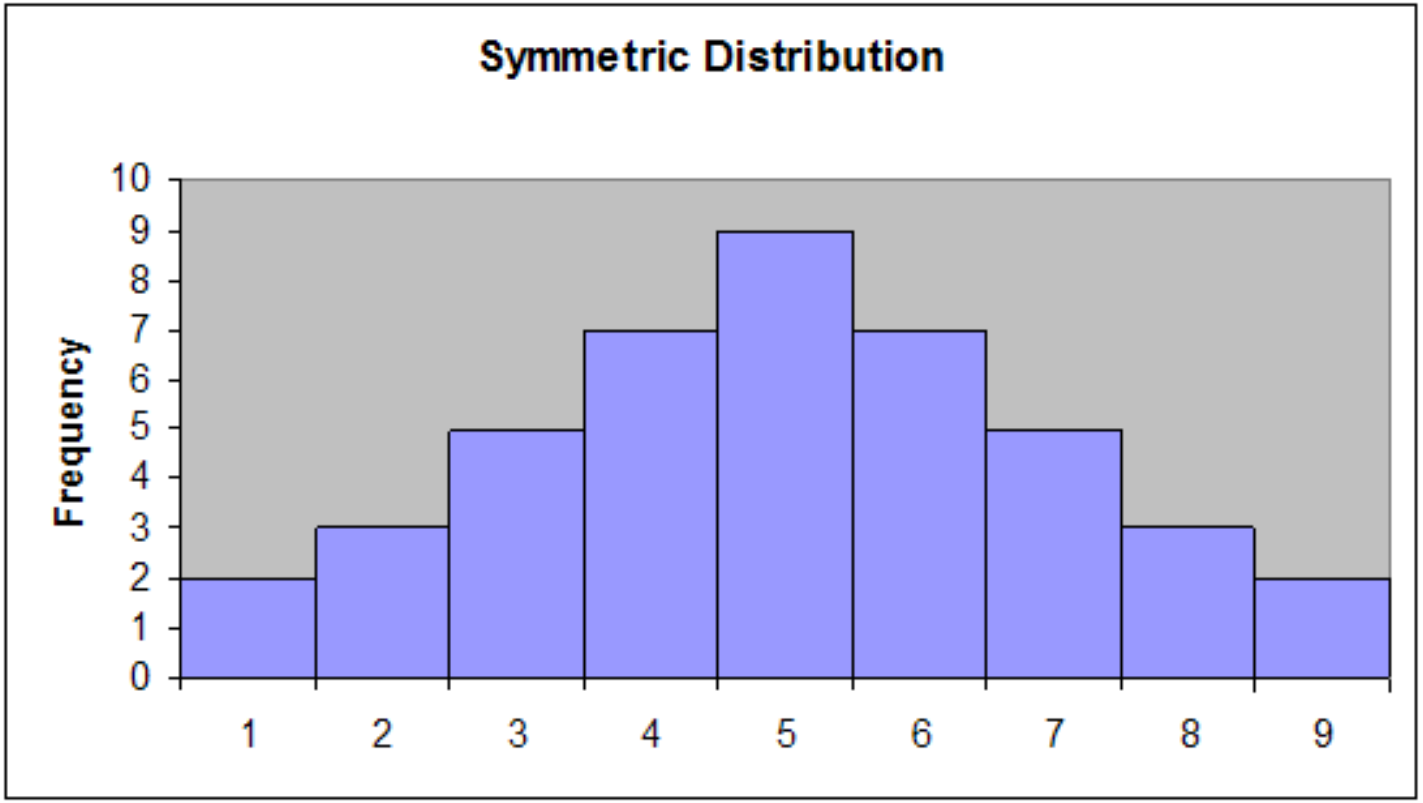
Histogram

The frequency histogram allows us to:

- describe the distribution of data,
- examine the symmetry/asymmetry, concentration/dispersion of the data,
- identify normal distribution (bell-shaped),
- determine the mode (unimodal, bimodal).

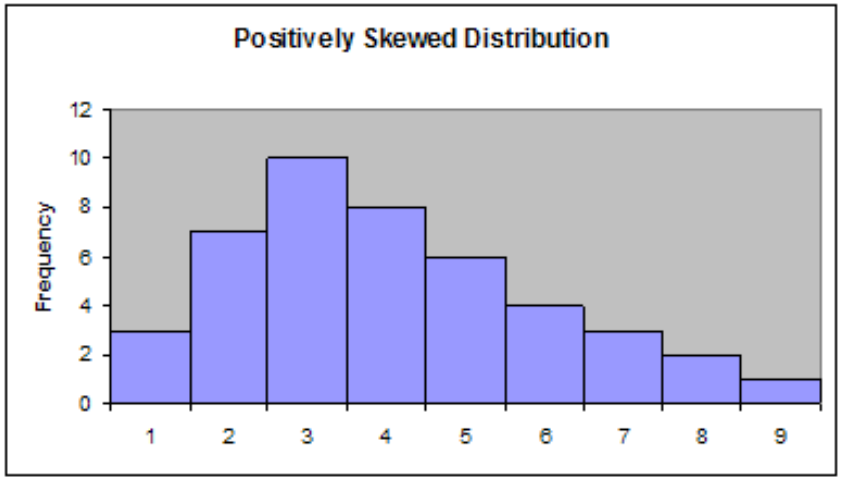
Shape of the distribution

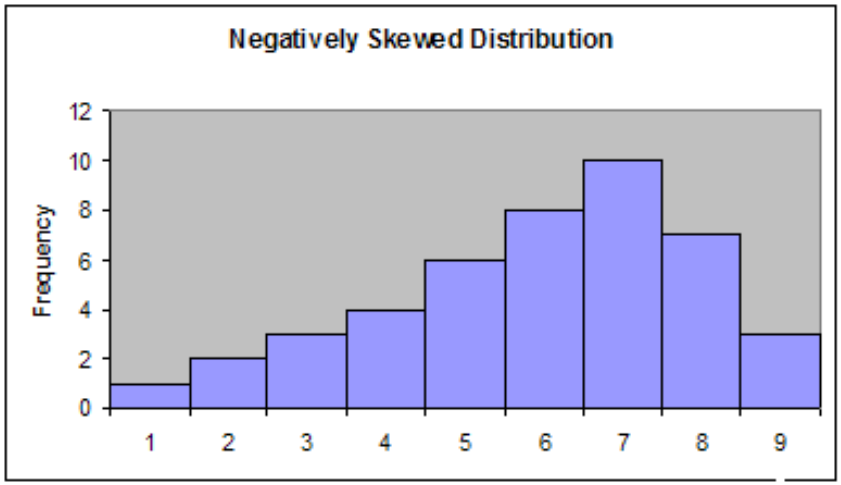
- The shape of the distribution is called symmetric if the observed values are balanced around the center.



Shape of the distribution

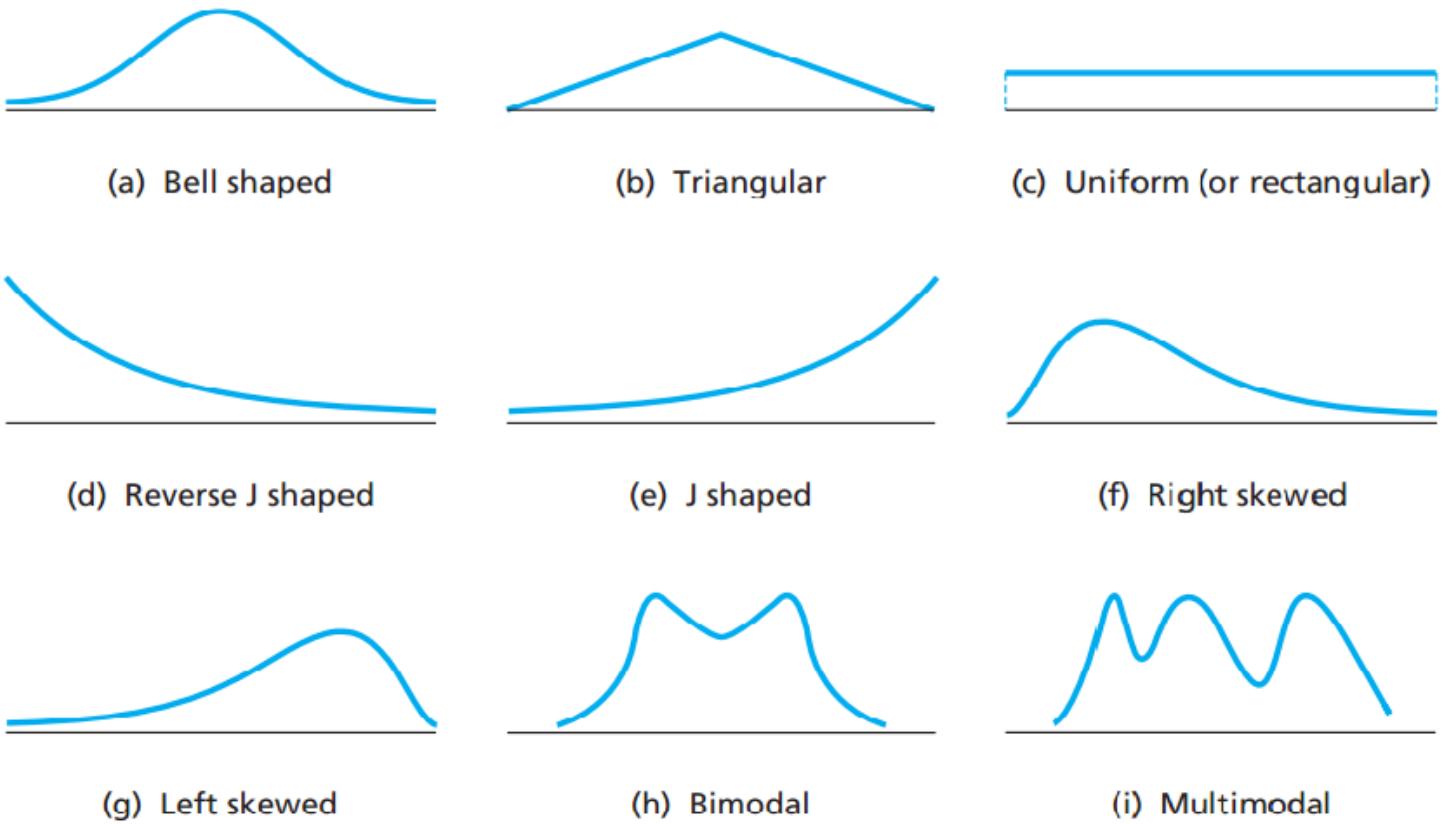
- The shape of a distribution is called skewed if the observed data is not distributed symmetrically around the center.





Shape of the distribution

- Use a histogram to identify the probability distribution of a random variable.



Stem-and-leaf plot

- It is a simple type of plot used to identify the distribution of a variable from a dataset.
- **Method:** sort data in ascending order, divide the sorted values into two parts: the first part consists of the leading digits (stem) and the second part is the trailing digit (leaf).

Stem-and-Leaf Plot - Example

Data in ordered array:

21, 24, 24, 26, 27, 27, 30, 32, 38, 41

- Here, use the 10's digit for the stem unit:
  - 21 is shown as
  - 38 is shown as

Stem	Leaf
2	1
3	8

Stem-and-leaf plot - example

Data in ordered array:

21, 24, 24, 26, 27, 27, 30, 32, 38, 41

- Completed stem-and-leaf diagram:

Stem	Leaves
2	1 4 4 6 7 7
3	0 2 8
4	1

Stem-and-leaf plot - example

- Using the 100's digit as the stem:
  - Round off the 10's digit to form the leaves
    - 613 would become
    - 776 would become
    - . . .
    - 1224 becomes

Stem	Leaf
6	1
7	8
12	2

Stem-and-leaf plot - examples

Example 1

The following data set describes the Mathematics exam scores (100-point scale) of 20 students in a class.

72	49	62	58	73	55	78	83	57	63
73	73	75	85	85	64	61	67	75	91

Construct the stem-and-leaf plot for the above data set.

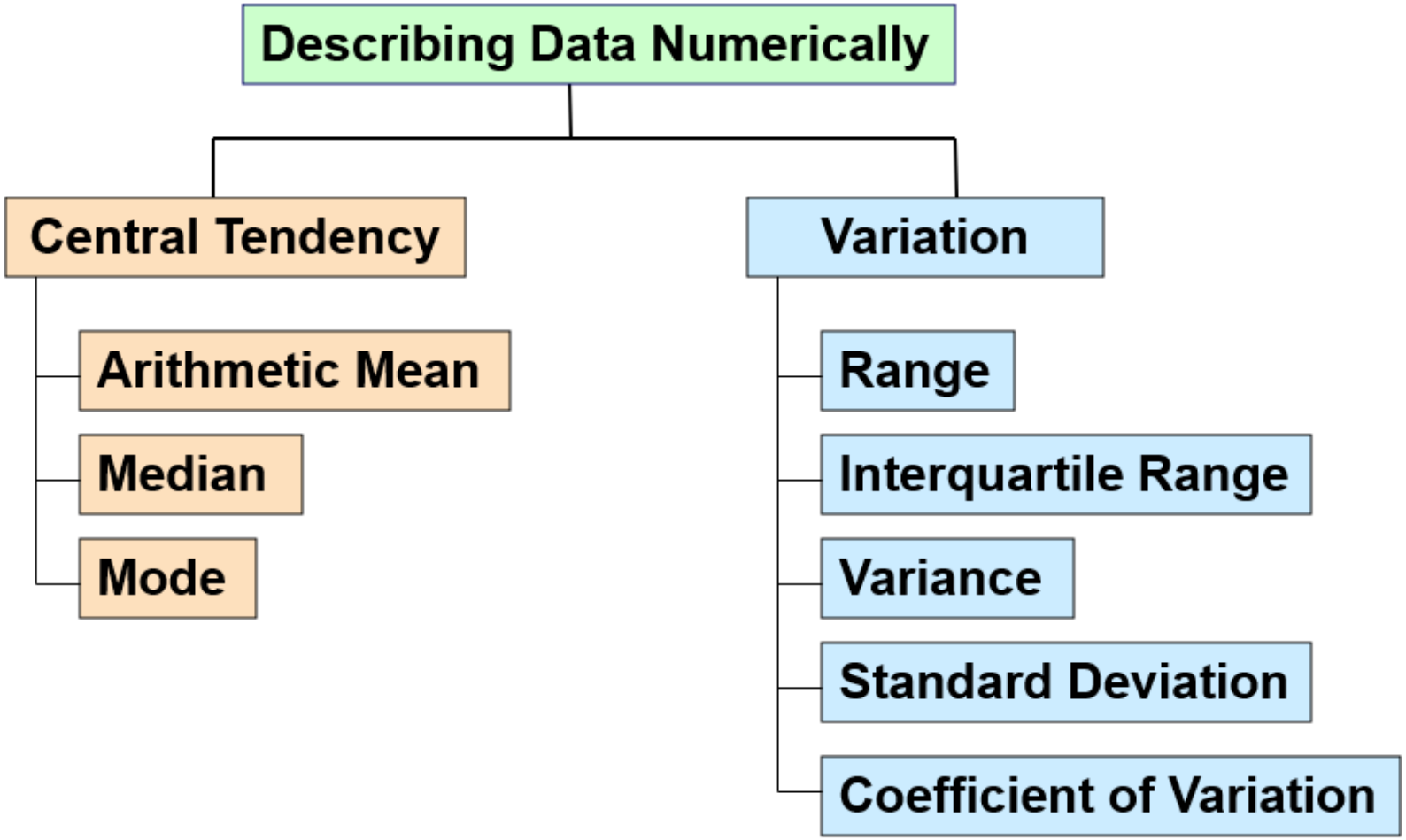
Example 2

The data set below shows the results of the long jump exam (Unit: m) of 10 students in a physical education course:

2.3	2.5	2.5	2.7	2.8	3.2	3.6	3.6	4.5	5.0

Construct the stem-and-leaf plot for the above data set.

Introduction



Measures of Central Tendency

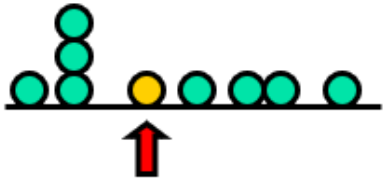
Central Tendency

Mean

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

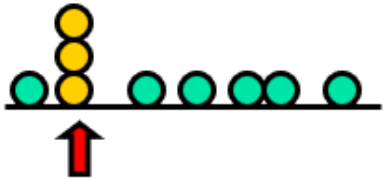
Arithmetic average

Median



Midpoint of ranked values

Mode



Most frequently observed value

Mean

- The mean is the most commonly used measure of the central value of data.
- For a population with N elements, the population mean is given by

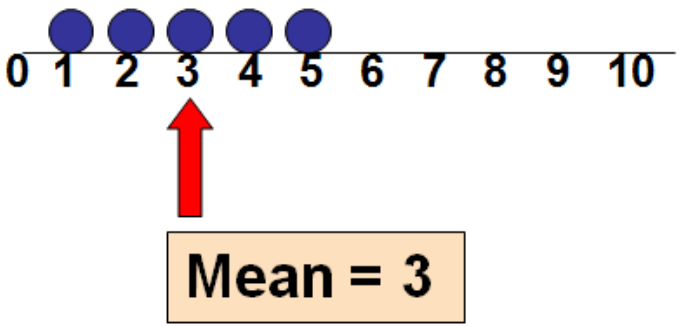
$$\mu = \frac{\sum_{i=1}^{N} x_i}{N} = \frac{x_1 + x_2 + \ldots + x_N}{N}.$$

- For a sample of size n, the sample mean is

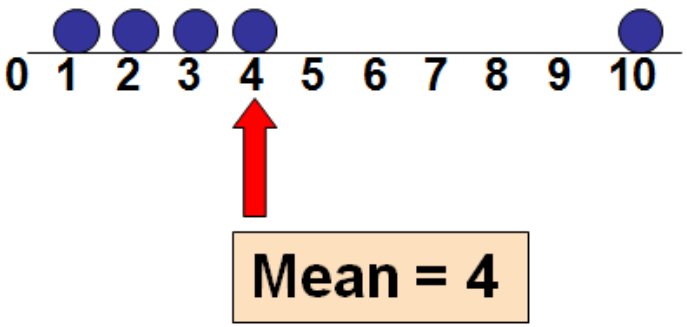
$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + \ldots + x_n}{n}.$$

Mean

- The mean is affected by outliers.



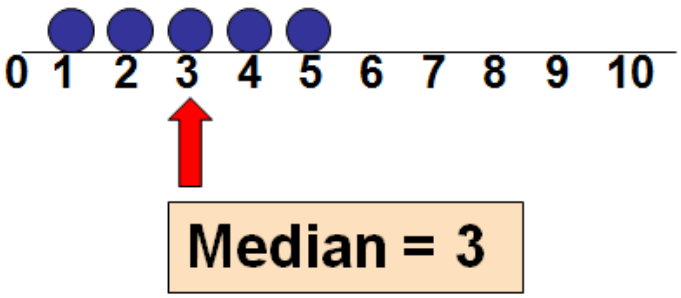
$$\frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

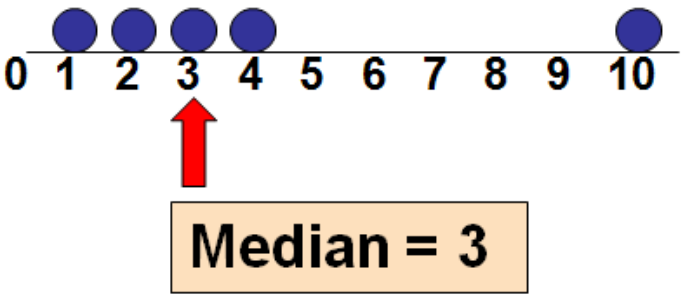


$$\frac{1+2+3+4+10}{5} = \frac{20}{5} = 4$$

Median

- In a dataset sorted in ascending order, the median is the "middle" value of the data (50% above, 50% below).
- The median is not affected by outliers.





Median

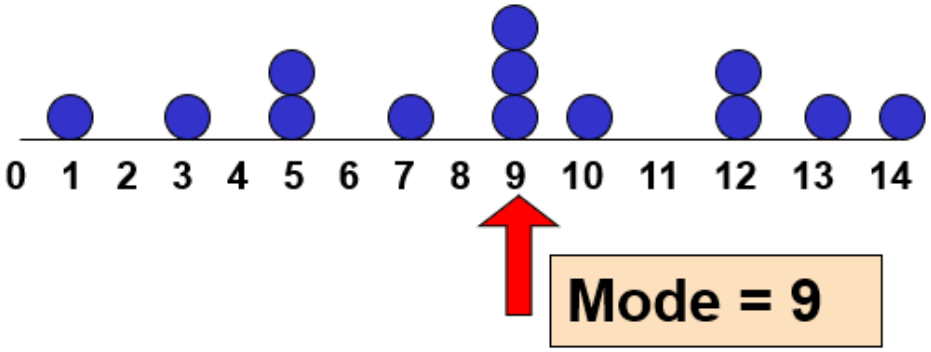
- Position of the median: sort the data in ascending order, let i be the position of the median

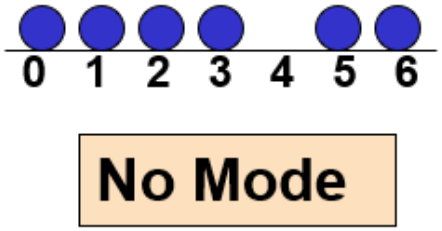
$$i = \frac{n+1}{2}$$

  + If i is an integer, median $= X_i$,
  + If i is not an integer, median $= \dfrac{X_{[i]} + X_{[i]+1}}{2}$, where $[i]$ is the integer part of i.

Mode

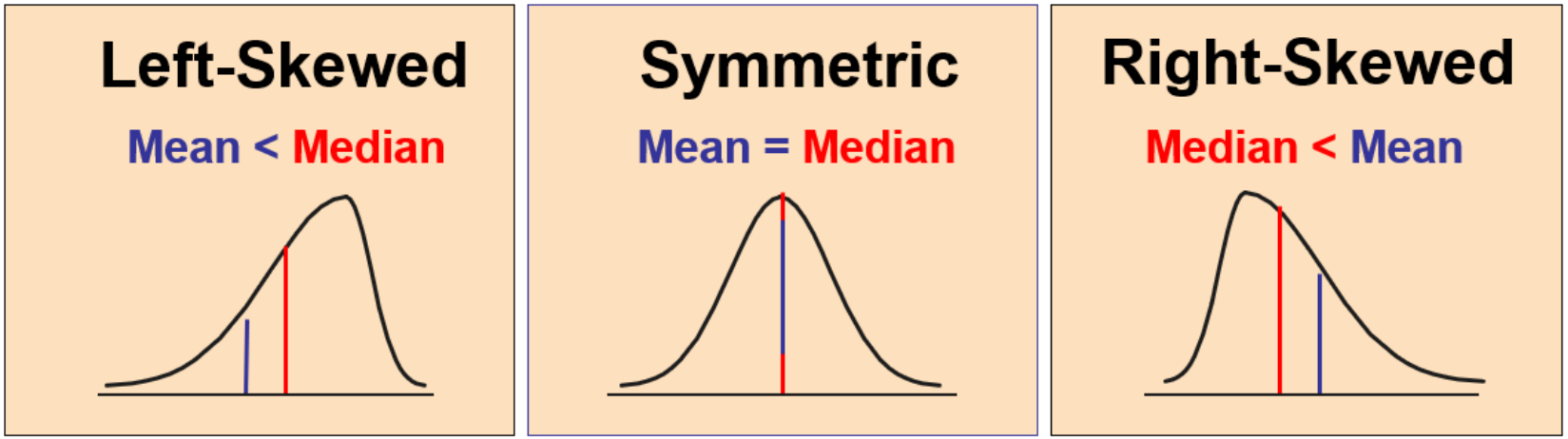
- Mode is a measure of the central tendency of data,
- is the value that occurs most frequently,
- is not affected by outliers,
- can be used for both qualitative and quantitative data,
- there may be multiple modes or no mode.





Which measure is the best?

- The mean is always used if outliers do not exist or after removing outliers.
- The median is often used if the dataset contains outliers or is highly skewed.
- The mode is often used to describe qualitative variables.
- The positions of the median and mean are affected by the shape of the distribution:



Measures of center - example

Example 3

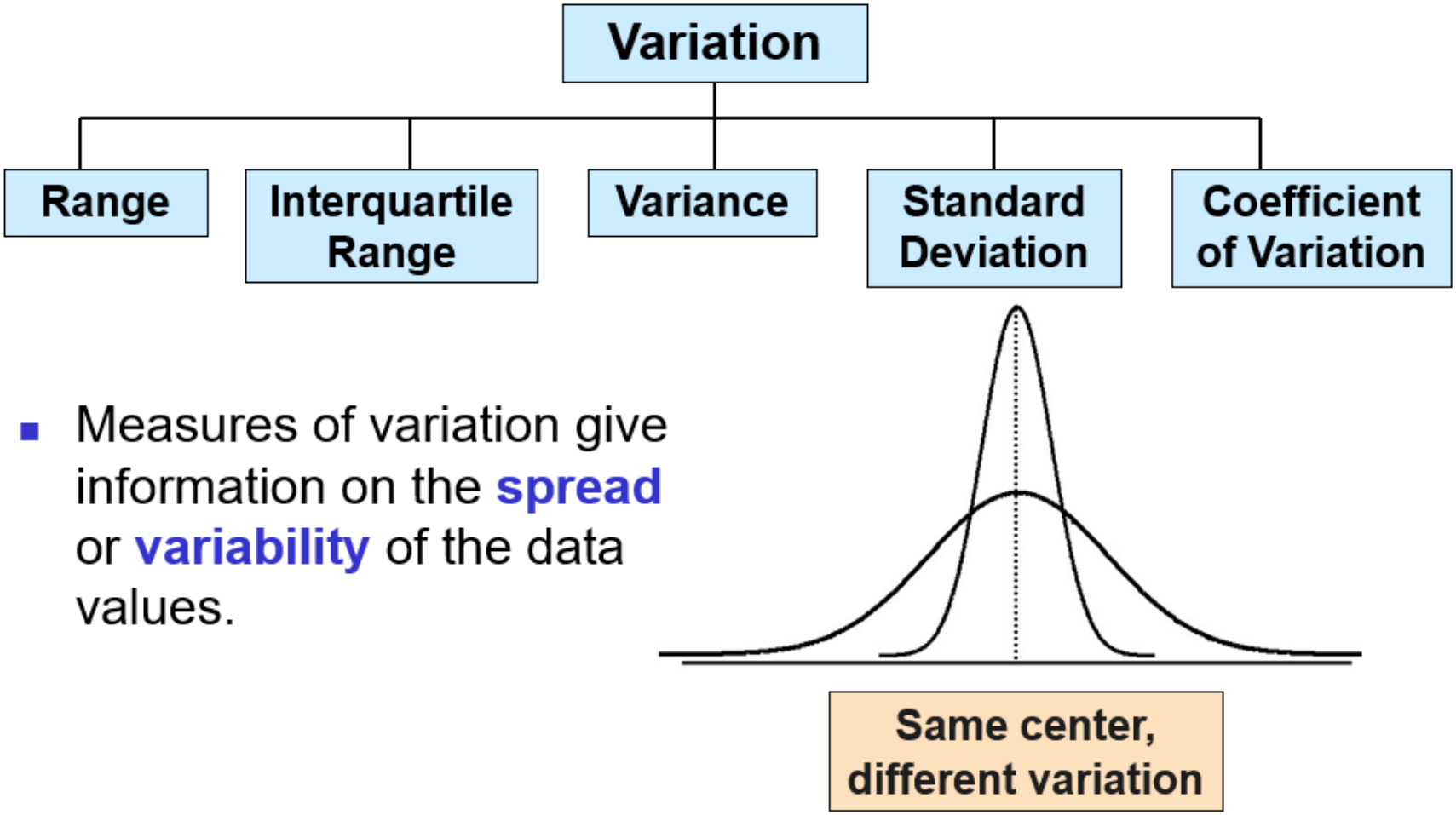
The following data set describes the Math exam results (100-point scale) of 20 students in a class.

72	49	62	58	73	55	78	83	57	63
73	73	75	85	85	64	61	67	75	91

a) *Construct a stem-and-leaf plot for the above data set.*

b) *Calculate the mean, median, and mode.*

Measures of Variation

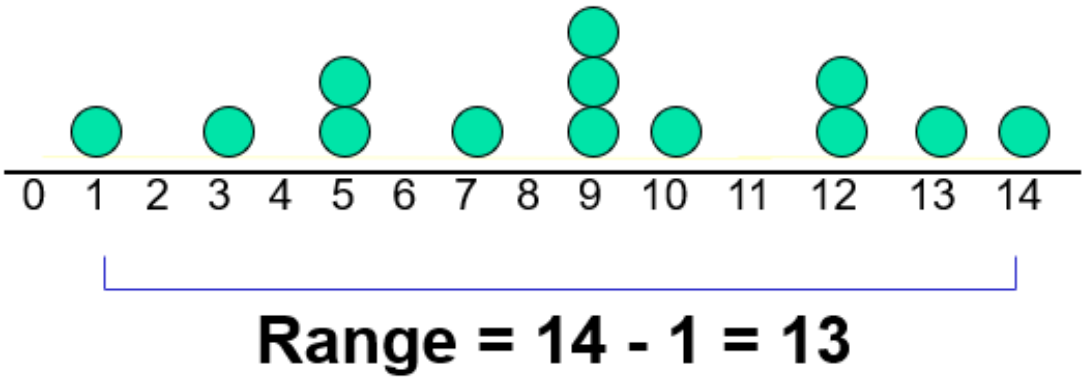


- Measures of variation give information on the **spread** or **variability** of the data values.

Range

- Range is the simplest measure of variation,
- It is the difference between the maximum and minimum values of the observed data

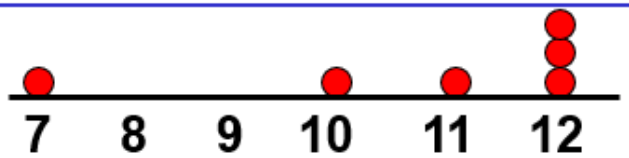
$$\text{Range} = X_{Max} - X_{Min}$$



Disadvantages of the range

- Ignores the way in which data are distributed

Range = 12 - 7 = 5



Range = 12 - 7 = 5

- Sensitive to outliers

1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,3,3,3,3,4,5

Range = 5 - 1 = 4

1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,2,3,3,3,3,4,120

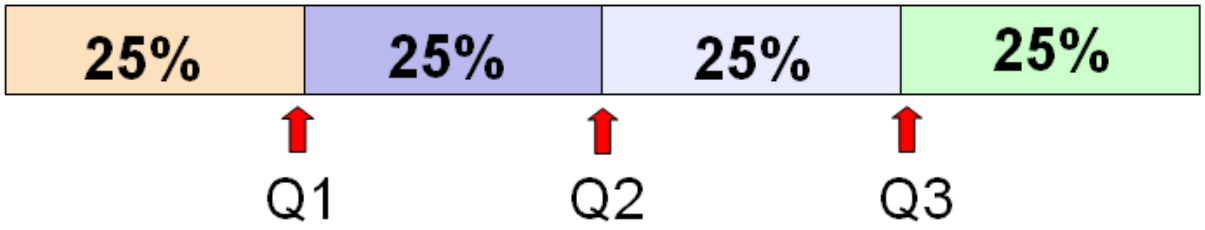
Range = 120 - 1 = 119

Inter-Quartile Range

- We can eliminate outliers by using the inter-quartile range (IQR) (also known as the midspread).
- The formula for the inter-quartile range:

$$IQR = Q_3 - Q_1$$

  where Q_1 is the first quartile (25% level) and Q_3 is the third quartile (75% level) of the data.
- The points Q_1, Q_2, and Q_3 are called the quartiles.



Formula for finding quartiles

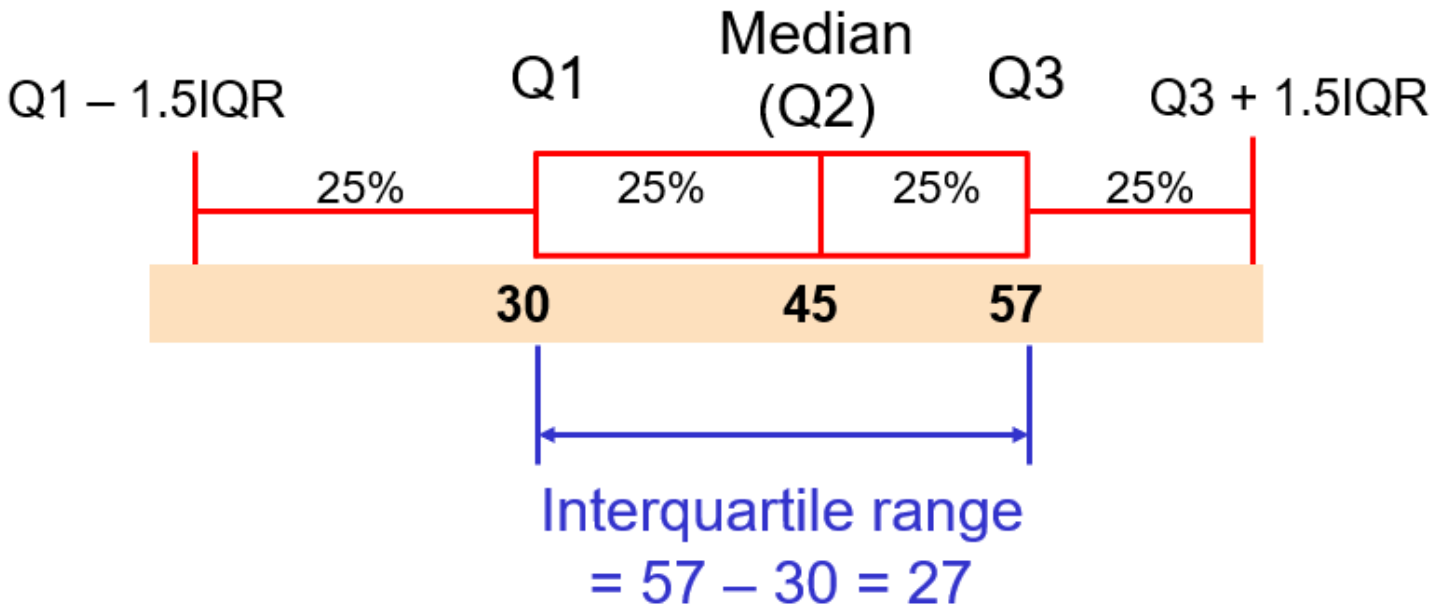
Arrange the data in ascending order, let Q_1, Q_2 (median), and Q_3 be the first, second, and third quartiles of the data, respectively. The positions of Q_1, Q_2, and Q_3 are determined as follows

$$\text{Position of first quartile} = 0.25(n+1)$$
$$\text{Position of second quartile} = 0.5(n+1)$$
$$\text{Position of third quartile} = 0.75(n+1)$$

where n is the number of observations.

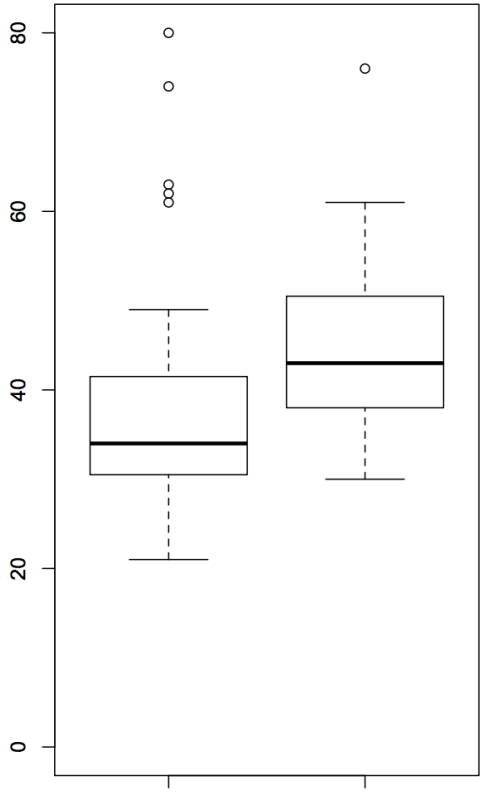
Box plot

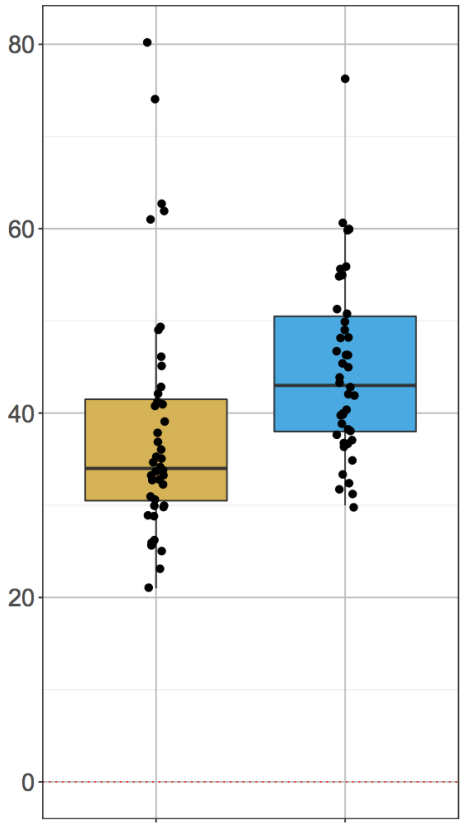
- To represent the interquartile range and outliers, we use a box plot:



Boxplot

- By plotting multiple boxplots of different datasets side-by-side, one can compare the dispersion and central values (mean/median) of these datasets.





Variance

- Variance is the average of squared deviations of values from the mean.
- Variance reflects the dispersion or variability of the data.
- Population variance

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

where N is the population size, μ is the population mean, and x_i is the i-th value of variable x.

- Sample variance

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

where $\bar{X}$ is the sample mean, n is the sample size, and X_i is the i-th value of variable X.

Standard Deviation

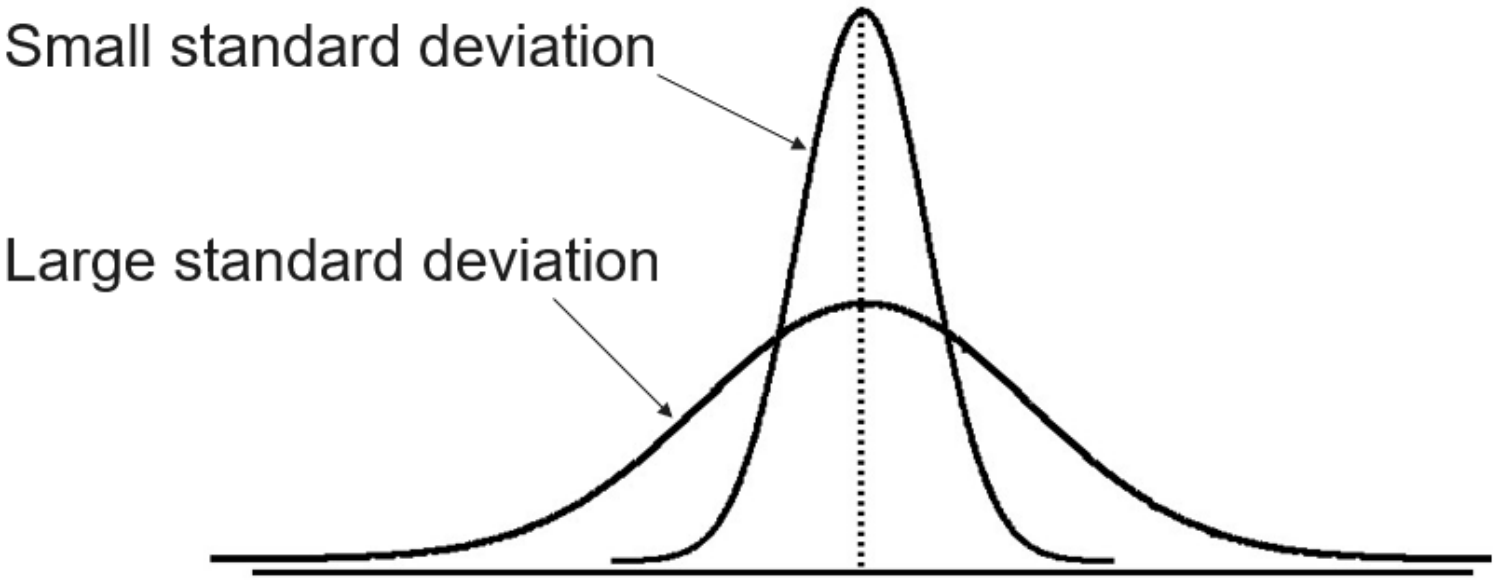
- Standard deviation is used to measure variation, representing the variation around the mean,
- Has the same unit of measurement as the original data.
- Population standard deviation, denoted by σ:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}.$$

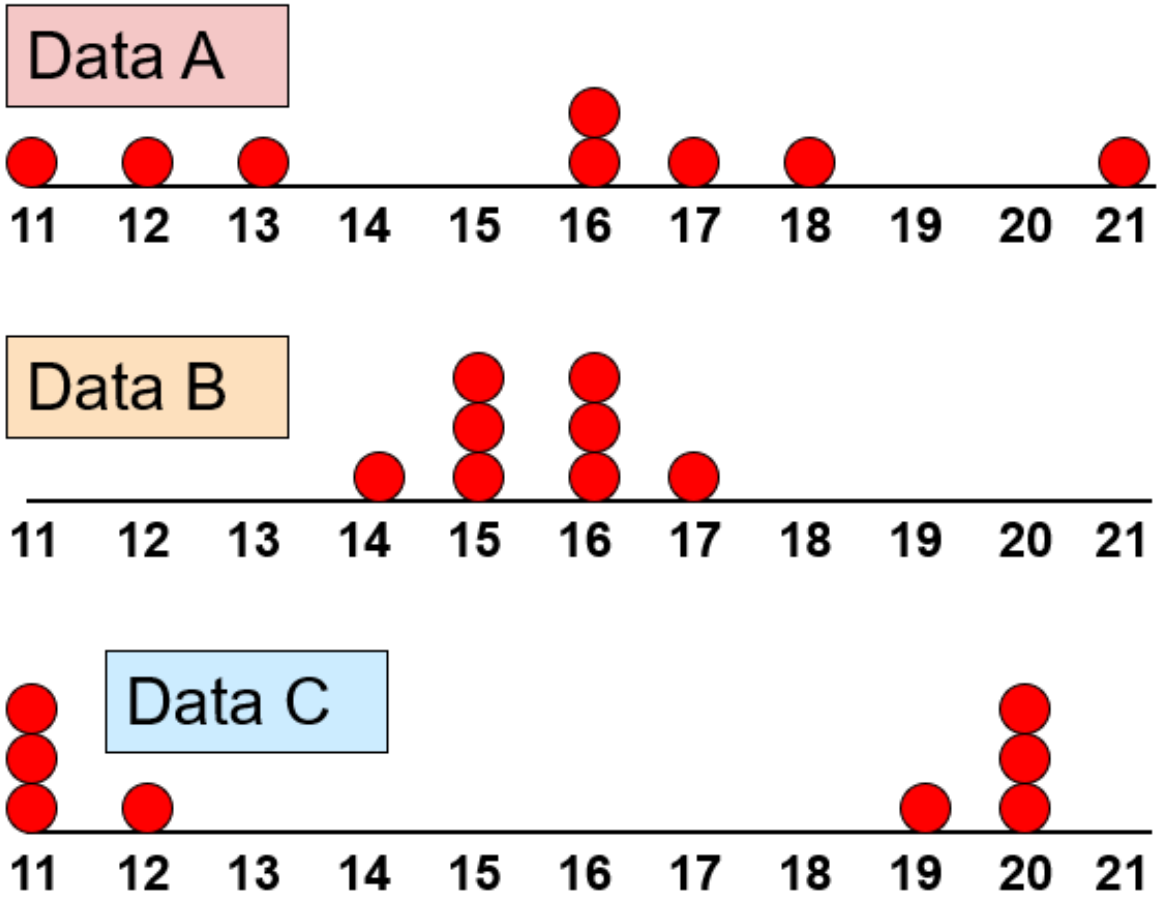
- Sample standard deviation,

$$S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}.$$

Measures of Variation



Comparing Data Variation using Standard Deviation



Mean = 15.5
s = 3.338

Mean = 15.5
s = 0.926

Mean = 15.5
s = 4.570

Coefficient of Variation

- Coefficient of Variation (CV) is used to compare the variability of two or more datasets, which may be measured in different units.
- It measures the relationship between variability and the mean.
- It is expressed in %.
- Formula:

$$CV = \frac{S}{\bar{X}} 100\%.$$

Comparing Coefficients of Variation

- Data set A:
  - Mean $\bar{x}_A = 50$
  - Standard deviation $s_A = 5$

$$CV_A = \frac{S_A}{\bar{x}_A} 100\% = \frac{5}{50} 100\% = 10\%.$$

- Data set B:
  - Mean $\bar{x}_B = 100$
  - Standard deviation $s_B = 5$

$$CV_B = \frac{S_B}{\bar{x}_B} 100\% = \frac{5}{100} 100\% = 5\%.$$

- Both data sets have the same standard deviation, but data set B shows less variability relative to its mean.

Measures of Variability – Example

Example 4

The following data set describes the Mathematics exam scores (out of 100) of 20 students in a class.

72	49	62	58	73	55	78	83	57	63
73	73	75	85	85	64	61	67	75	91

a) *Find* Q_1, Q_2, Q_3.

b) *Draw a boxplot for the data set above.*

c) *Compute the variance and the standard deviation.*

Chebyshev's Theorem

Theorem 1 (Chebyshev)

For any population with mean μ and standard deviation σ, and for $k > 1$, the percentage of observations that lie within the interval

$$[\mu \pm k\sigma]$$

is at least

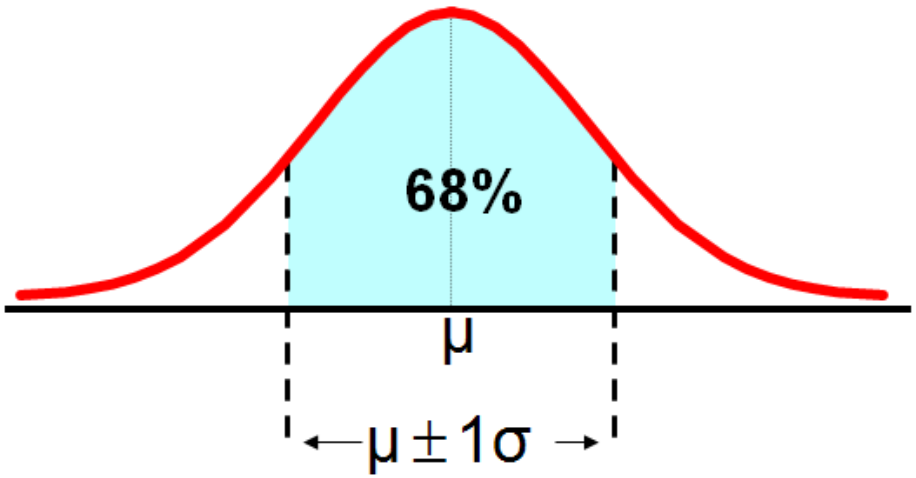
$$100\left[1 - \frac{1}{k^2}\right]\%.$$

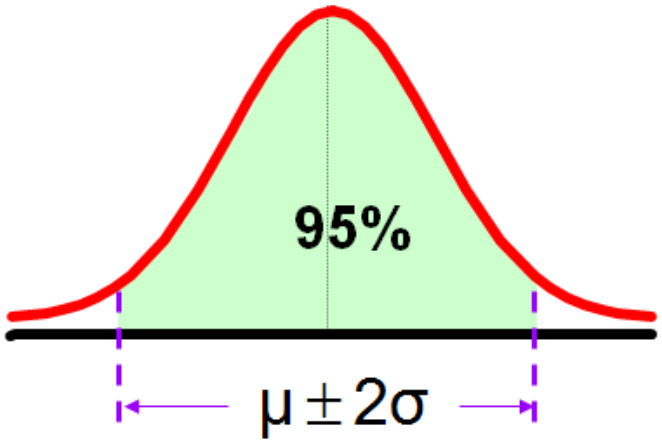
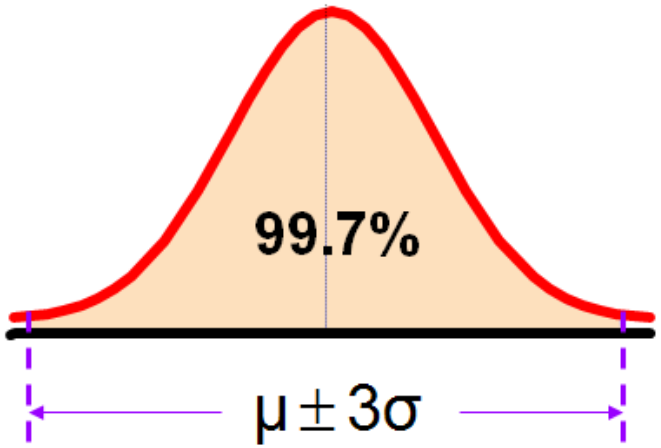
- Example:

At least		within
$(1 - 1/1^2) = 0\%$	.....	$k=1$ $(\mu \pm 1\sigma)$
$(1 - 1/2^2) = 75\%$	.....	$k=2$ $(\mu \pm 2\sigma)$
$(1 - 1/3^2) = 89\%$	.....	$k=3$ $(\mu \pm 3\sigma)$

Empirical Rule

- Empirical Rule: if the data follow a normal distribution (or are approximately normal), then the interval
  - $\mu \pm 1\sigma$ contains approximately 68% of the data values in the sample or population.



Empirical Rule

- $\mu \pm 2\sigma$ contains approximately 95% of the data values of the sample or population.
- $\mu \pm 3\sigma$ contains approximately 99.7% of the data values of the sample or population.